

STARTUPHUB: PITCH-TO-STARTUP

AI-Powered Global Venture Ecosystem & Real-Time Pitch Matchmaking Platform

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1. ABSTRACT

Early-stage startup fundraising, evaluation, and recruitment face severe structural friction, characterized by static PDF decks, low-response cold outreach, and fragmented tools. This project introduces StartupHub (Pitch-to-Startup), an end-to-end web platform unifying founders, angel/VC investors, and specialized tech talent.

The system delivers a gesture-controlled Dealflow Swiper with interactive drag-physics, dynamic visual stamps, and Web Audio API synthesis. Startups feature an interactive 7-Slide Pitch Deck Studio, an Autonomous AI Due Diligence Scorecard, a 60-second elevator audio player with synced teleprompter subtitles, and a 1-click Term Sheet generator with client-side PDF export. The platform also integrates an encrypted 1-on-1 WebRTC Video Pitch Room with deck screen sharing and a Talent Marketplace with an automated ATS resume optimizer.

2. PROBLEM STATEMENT & OBJECTIVES

- **High Screening Inefficiency:** Investors spend up to 4 minutes per deck scanning unstructured PDFs, causing fatal pipeline bottlenecks.
- **Zero Interactive Presence:** Traditional email decks cannot convey founder passion, interactive unit economics, or live product demos.
- **Fragmented Ecosystem:** Teams rely on disparate vendors (DocSend, Zoom, AngelList, LinkedIn), fracturing fundraising telemetry.
- **Primary Goal:** Engineer a unified platform supporting gesture dealflow curation, standardized interactive pitch decks, instant WebRTC meetings, and AI candidate matching.

3. SYSTEM ARCHITECTURE & DESIGN

StartupHub implements a decoupled, modern multi-tier architecture:

- **Frontend Tier:** Constructed with React 18, Vite, and Tailwind CSS. Features custom mouse/touch drag physics, jsPDF document generation, and Web Audio API oscillators for haptic feedback.
- **API Gateway & Security:** Spring Security 6 filter chain with stateless JSON Web Token (JWT) verification, CORS policies, and OAuth2 password challenges.
- **Application Logic:** Spring Boot 3.2.3 running on Java 17, exposing REST endpoints for startup dealflow and STOMP WebSocket (/ws) for WebRTC signaling.
- **Data Storage:** Persistent H2 file database running in PostgreSQL compatibility mode, pre-seeded with 18 international startups across 6 global hubs.

4. CORE MODULE SPECIFICATIONS

- **AI Dealflow Swiper:** Supports left-drag (Pass), right-drag (Interested), and up-drag (Super-Like) with real-time stamp opacity modulation and undo/rewind capability.
- **7-Slide Pitch Deck Studio:** Seven structured slides: Executive Summary, Problem, Proprietary Solution Moat, TAM/SAM/SOM, Unit Economics, Team/Cap Table, and 18-Month Use of Funds.
- **Founder Elevator Pitch Dock:** 60-second animated audio visualizer waveform synced to chapters (Hook, Breakthrough, Traction).
- **WebRTC 1-on-1 Video Pitch Room:** Direct peer-to-peer audio/video streaming with getDisplayMedia screen-sharing and real-time captions.
- **ATS Resume Optimizer:** Analyzes applicant resumes against job specs, producing compatibility scores and single-column ATS-ready resumes.

5. DATABASE SCHEMA & ENTITY DEFINITIONS

The relational schema models the multi-sided startup ecosystem:

- **StartupEntity:** Stores UUID, name, valuation, ARR, MRR, growth %, city, country, KYC status, stakeholdersJson, and fundingLedgerJson.
- **JobApplicationEntity:** Tracks applicant profiles, target startup, role type, compensation range, and ATS match score (0-100%).
- **MarketingCampaignEntity:** Maintains AI-generated omni-channel campaigns across LinkedIn, Twitter/X, and TechCrunch.

6. RESULTS, VERIFICATION & DEPLOYMENT

The application was successfully compiled, packaged, and verified:

- **Backend Service:** Spring Boot embedded Tomcat operational on port 8080. Hibernate DDL schema initialized in 2.4s.
- **Frontend Client:** Vite v5.4 dev server running on port 3000 with clean HMR proxying. Production bundle compiled in 9.27s.
- **Deployment Ready:** Configured with vercel.json SPA rewrites and committed to GitHub (<https://github.com/Imran8624/Pitch-to-Startup>) on branch main.

7. CONCLUSION

StartupHub successfully demonstrates how gamified UX paradigms, standardized pitch deck slides, autonomous AI diligence, and WebRTC streaming can coalesce into an efficient, unified venture platform, substantially accelerating early-stage startup matchmaking.